Making NDFU bottom-up: the unified adoption rule and its two variants

NDFU internal design note

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Status: design document — no changes to the engine have been made.
Companion scripts: `10_unified_model_sweep.R`, `12_unified_premium_sweep.R`

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1 Why this document

The Social Networks special-issue rejection (Bianchi's follow-up email) said:

> *"the underdevelopment of the behavioural model assumed in your ABM. While it can certainly be considered an agent-based model, the behaviour of the agents is relatively*

abstract, and the experimental design appears to be governed by macro-level manipulation rather than bottom-up dynamics."

Two pieces of the current model are exposed to that critique:

1. **The network threshold τ_i is exogenous.** It is drawn from $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$, truncated to $[0, 1]$, *independently of the agent's own measured preference* q_i. The sweep over $(\tau_{\text{mean}}, \tau_{\text{sd}})$ is exactly the "macro-level manipulation" Bianchi points at: a population-level dial with no micro-foundation in the survey data.
2. **The two channels are bolted together.** Rational adoption ($q_i \leq \Gamma$) and social adoption ($\tau_i \leq \tilde{E}_i$) are two separate rules joined by an OR. A referee can ask why an agent runs two unrelated decision procedures.

Tur, Zeppini & Frenken (2024, *Social Networks* 76:12–21) — published in our target journal — has a *single* micro rule (a utility comparison, where social reinforcement enters the utility calculus), but **no social-distance sensitivity in influence**: every adopting neighbor reinforces equally. NDFU's distinctive ingredient is exactly that sensitivity (Centola 2011). The question is therefore:

> Can we keep NDFU's selective (distance-sensitized) influence while adopting a Tur-style single utility-based decision rule — making the model bottom-up without giving up its identity?

Answer: yes. The instrument is a single **unified adoption rule** (§4), which admits two operationalizations — a *linear-fractional* variant and a *multiplicative γ-variant* that nests Tur (2024) exactly. Their pros and cons are compared in detail in §5.

2 Notation and the current NDFU model (baseline)

Symbol	Meaning	Source
$\Gamma \in [0, 1]$	Intrinsic Utility Level (IUL) of the behavior	swept
$q_i \in [0, 1]$	Minimum Utility Requirement (MUR): intrinsic aversion	**empirical** (GSS/ATP items)
$\tau_i \in [0, 1]$	network threshold (fraction of influential adopters needed)	**exogenous** $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$
$d_{ij} \in [0, 1]$	Gower distance in Blau space (age, educ, race, relig, sex)	**empirical**
$h \in [0, 1]$	Maximum Social Proximity (MSP): tolerance to dissimilarity	swept (macro)
$a_j \in \{0, 1\}$	adoption state of j	dynamic
X_{ij}	adjacency	imputed ERGM networks

Current rules (as in the paper).

- Rational channel: adopt if $q_i \leq \Gamma$ (Eq. 1 of the paper).
- Social channel: adopt if $\tau_i \leq \tilde{E}_i$, with selective effective exposure

$$\tilde{E}_i \;=\; \frac{\sum_{j \neq i} X_{ij}\, \tilde{a}_j}{\sum_{j \neq i} X_{ij}}, \qquad \tilde{a}_j = 1 \;\iff\; a_j = 1 \wedge U_{ij} \leq \sigma(h - d_{ij}), \tag{1}$$

where $U_{ij} \sim \text{Unif}(0, 1)$ is redrawn per step and $\sigma(z) = \left(1 + e^{-z/T}\right)^{-1}$ with $T = 0.02$ in the implementation (a near-hard gate at $d_{ij} = h$; see `scripts/04_GSS_diffusion_sims.R`, line 304).

Implementation nuance worth knowing (it matters for the comparison with Tur): in the code, the rational channel is implemented by setting the node's threshold to 10^{-6}, so a rational agent still needs **one influential contact** (exposure > 0) before adopting. The engine is therefore already a word-of-mouth model for *awareness*: low-q agents behave as simple contagion, not as spontaneous ($\varphi = 0$) adopters. The paper's Eq. 1 (adoption at exposure 0) is the idealization; the code is contact-gated.

3 Tur (2024) mechanism, for reference

- Word-of-mouth **percolation**: agents must be *informed* through the network before they can decide.
- Single decision rule: adopt when the innovation's intrinsic utility ξ meets the agent's minimum utility requirement.
- **Social reinforcement modifies the requirement** — exact published form (confirmed against both PDFs):

$$u_t^i \;=\; u_0^i \cdot \left(\frac{1}{a_t^i}\right)^{\gamma}, \tag{2}$$

  where u_0^i is the initial minimum utility requirement, $a_t^i \in \mathbb{N}$ the number of **adopting neighbors** of i at time t (a state count, not received recommendations; the rule is evaluated only once informed, so $a \geq 1$), and $\gamma \geq 0$ the social-reinforcement strength. The lowering is **multiplicative** (power-law): each doubling of adopting neighbors divides the requirement by the same factor 2^{γ}. With $\gamma = 0$ the model reduces to pure social percolation. The threshold-like behavior emerges from this lowering, not from a separate threshold parameter.
- Homophily ρ is purely **structural pre-wiring**: agents with similar u are placed together *before* diffusion. Influence itself is **distance-blind**: any adopting neighbor reinforces equally.
- Preferences are artificial distributions (Beta(α, β) family in 2018; uniform in 2024); networks are Watts–Strogatz with $N = 10{,}000$ and $k = 4$ neighbors on average.

NDFU vs. Tur, in one line: **Tur has the single bottom-up rule but uniform influence; NDFU has selective influence but a two-channel rule with an exogenous τ.** Each has what the other lacks.

4 The unified additive-utility rule

4.1 The rule

Replace both channels by one utility comparison per agent per step:

$$\text{adopt}_i(t) = 1 \iff \Gamma + \lambda\, \tilde{E}_i(t) \;\geq\; q_i. \tag{3}$$

- $\tilde{E}_i(t)$ is **exactly the current selective effective exposure** (1), unchanged. This is where social distance enters: only adopters passing the stochastic proximity gate $\sigma(h - d_{ij})$ count.
- $\lambda > 0$ is the **social-proof coupling**: how much situational utility a fully-adopted, fully-similar neighborhood adds, in the same units as Γ and q_i.

4.2 Two algebraically identical framings (choose the second)

1. **Requirement-lowering (Tur's framing):** $q_i^{\text{eff}}(t) = q_i - \lambda \tilde{E}_i(t)$; adopt when $\Gamma \geq q_i^{\text{eff}}(t)$. Preferences are *modified* during diffusion.

2. **Situational-utility (recommended framing):** $U_i(t) = \Gamma + \lambda \tilde{E}_i(t)$; adopt when $U_i(t) \geq q_i$. The preference q_i **never changes**; what changes is the utility the situation offers (peers add legitimacy, lower uncertainty, network externalities). This is literally Granovetter's threshold as a *situated cost–benefit crossing* (1978, p. 1422), and it preserves NDFU's declared contrast with Tur ("preferences are not modified during diffusion", `theoretical_reviewed.md`, entry 7).

4.3 The derived threshold — the two old channels fall out as limit cases

Rearranging (3), adoption requires $\tilde{E}_i(t) \geq \tau_i^*$ with the **derived threshold**

$$\tau_i^* \;=\; \max\left\{0,\; \frac{q_i - \Gamma}{\lambda}\right\}. \tag{4}$$

- $q_i \leq \Gamma \;\Rightarrow\; \tau_i^* = 0$: adopts at first influential contact — **the current rational channel** (Granovetter's low-threshold "instigator", exactly as currently implemented with $\tau = 10^{-6}$).
- $q_i > \Gamma \;\Rightarrow\; \tau_i^* > 0$: needs selective exposure proportional to the **utility deficit** — the social channel, now derived instead of postulated.
- $q_i > \Gamma + \lambda \;\Rightarrow\; \tau_i^* > 1$: **can never adopt** ($\tilde{E} \leq 1$) — a "structurally immune" agent. When exactly this occurs is worked out in §5.1.

4.4 What this buys and what it costs

Buys.

- **Bottom-up microfoundation.** One decision rule, all heterogeneity empirical: τ_i^* is a deterministic function of the *measured* q_i and the product (Γ, λ). The arbitrary $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$ disappears. This is the direct answer to Bianchi.
- **Behavioral realism as a bonus:** resistance to peer influence is now positively correlated with intrinsic aversion (the same latent trait drives both) — defensible, and testable in principle.
- **Nesting.** Setting the gate to $w \equiv 1$ ($h \to \infty$, distance-blind) removes the distance selectivity and leaves a Tur-style (distance-blind) reinforcement model; setting $\lambda \to 0$ recovers a pure percolation-of-awareness model. Tur (2024)'s *exact* published form (2) is recovered by the γ-variant of the rule — see §4.6. NDFU's selectivity remains the *identified added ingredient*, cleanly positioned against a paper published in the target journal.
- **Implementability.** τ_i^* is static given (Γ, λ), so the existing `netdiffuseR` threshold machinery runs it as-is: `threshold.dist = pmax(0, (q - Γ)/λ)` replaces the Normal draw + the 10^{-6} hack. No engine changes needed.

Costs / changes.

- The rational/social decomposition becomes **ex-post bookkeeping** (adopters with $q_i \leq \Gamma$ vs. $q_i > \Gamma$) rather than two mechanisms — same tally we already compute, but the interpretation shifts.
- τ_{sd} disappears as a free dial; its large positive effect on cascades (heterogeneity creates susceptible minorities) is now carried by the *empirical* dispersion of q_i, scaled by $1/\lambda$. λ replaces $(\tau_{\text{mean}}, \tau_{\text{sd}})$ — the parameter space shrinks from 16 (4×4) threshold configurations to a 1-D λ sweep.
- The (IUL × MSP) heatmaps will not be identical: thresholds now co-vary with q. Whether the phase structure survives is an empirical question — answered: it does (see §7).

4.5 Where social distance enters (explicitly)

Only inside $\tilde{E}_i$, via the per-contact stochastic gate:

$$\Pr\big(j\text{'s adoption counts for } i\big) \;=\; \sigma\!\left(\frac{h - d_{ij}}{T}\right), \qquad T = 0.02. \tag{5}$$

Identical to the current model. The unified rule changes the *decision rule*, not the *influence filter*.

4.6 Recovering Tur (2024) exactly — the nesting, step by step

Tur's published rule (exact form, §3): an agent that has been *informed* through the network adopts when the product's utility meets its *current* requirement,

$$\xi \;\geq\; u_t^i, \qquad u_t^i = u_0^i \cdot \left(\frac{1}{a_t^i}\right)^{\gamma}, \tag{6}$$

with u_0^i the initial minimum utility requirement, $a_t^i \geq 1$ the number of adopting neighbors of i at time t (the rule is only evaluated once informed, so $a \geq 1$ and the expression is well-defined), and $\gamma \geq 0$ the social-reinforcement strength. $\gamma = 0$ gives $u_t = u_0$: pure social percolation (Solomon & Weisbuch 2000), Tur's own baseline.

Log-linearization — Tur IS an additive requirement-lowering rule, in logs. Taking logarithms (all quantities positive):

$$\ln \xi \;\geq\; \ln u_0^i \;-\; \gamma \,\ln a_t^i. \tag{7}$$

So the published reinforcement is additive with the concave gain $g(a) = \ln a$. This dictates the right unified rule: it must be additive in *log-utility*, i.e. **multiplicative in levels**.

The Tur-exact unified rule (γ-form). Define the distance-sensitized adopting *count*

$$\tilde{A}_i(t) \;=\; \sum_j X_{ij}\, \tilde{a}_j(t), \qquad \tilde{a}_j = 1 \;\iff\; a_j = 1 \,\wedge\, U_{ij} \leq \sigma\!\left(\frac{h - d_{ij}}{T}\right), \tag{8}$$

and let social proof *multiply* the situational utility:

$$\boxed{\text{adopt}_i(t) \;\iff\; \Gamma \cdot \big(\tilde{A}_i(t)\big)^{\gamma} \;\geq\; q_i} \qquad \text{(evaluated when } \tilde{A}_i \geq 1\text{)}. \tag{9}$$

Equivalently, in Tur's own requirement-lowering form: $q_{i,t} = q_i \cdot (1/\tilde{A}_i(t))^{\gamma}$, adopt when $\Gamma \geq q_{i,t}$. The preference q_i is fixed; what the similar adopters change is the *situational* requirement — the same framing as §4.2 (but see §5.5 for the substantive difference this form does carry).

Recovery, in one step. Switch off the proximity gate ($h \geq 1 + 5T \Rightarrow \sigma \to 1 \Rightarrow \tilde{a}_j = a_j$, so $\tilde{A}_i(t) = a_t^i$), and rule (9) reads

$$\Gamma \cdot (a_t^i)^{\gamma} \geq q_i \qquad \Longleftrightarrow \qquad \xi \;\geq\; u_0^i \cdot \left(\frac{1}{a_t^i}\right)^{\gamma} \tag{10}$$

under the identification

$$\boxed{\xi = \Gamma, \qquad u_0^i = q_i, \qquad \text{same } \gamma\,.} \tag{11}$$

The recovery is **exact on any graph**: Tur's reinforcement is a pure count, and the γ-form uses the count directly — no degree normalization is involved, so no degree-regularity caveat is needed (unlike the linear-fractional variant, which normalizes by degree and matches Tur's count reading only on degree-regular substrates).

Awareness / percolation layer (unchanged): the engine's contact-gating (the $\tau = 10^{-6}$ device, §2) means the rule is only evaluated at $\tilde{A}_i \geq 1$ — precisely Tur's "informed" condition with per-contact transmission probability 1; a probability $p < 1$ is emulated by a constant-acceptance gate ($\sigma \equiv p$).

Derived threshold (count form). Rearranging (9):

$$\text{adopt}_i \iff \tilde{A}_i(t) \geq c_i^*, \qquad c_i^* = \left(\frac{q_i}{\Gamma}\right)^{1/\gamma}. \tag{12}$$

- $q_i \leq \Gamma \Rightarrow c_i^* \leq 1$: adopts at the first influential contact — the rational channel, exactly as in the linear variant (§4.3).
- $q_i > \Gamma \Rightarrow$ needs c_i^* similar adopters; heterogeneity 100% empirical (from q_i), with γ as the single coupling.
- **Degree-dependent immunity**: agent i is out of reach whenever $c_i^* > k_i$ (its degree) — high-aversion, *low-degree* agents can never be socially converted (see §5.1).

What does NOT nest (honesty box).

- Tur's homophily ρ sorts agents by the requirement u *itself*; NDFU's imputation sorts by demographics, and q_i correlates with demographics only partially. The imputed networks are therefore an *empirically disciplined, weaker* version of Tur's preference-sorting. The nesting claim concerns the **adoption rule**; the network is an exogenous input in both models.

One-line summary. With the proximity gate switched off, the γ-form **is** Tur (2024) — same functional form, same γ, with $\xi = \Gamma$ and $u_0 = q$ — on any topology; NDFU's added ingredient is exactly one: the distance gate deciding *which* adopters count in $\tilde{A}_i$.

5 Pros and cons of the two operationalizations

We now have two members of one family on the table:

$$\textbf{Linear-fractional:} \quad \Gamma + \lambda \tilde{E}_i(t) \geq q_i \quad \text{(additive in the adopted } \textit{fraction}\text{)} \tag{13}$$

$$\gamma\textbf{-form (Tur-exact):} \quad \Gamma \cdot \tilde{A}_i(t)^{\gamma} \geq q_i \quad \text{(multiplicative in the adopted } \textit{count}\text{)} \tag{14}$$

This section works through the five questions that decide between them, pedagogically and one at a time, and closes with a verdict table.

5.1 When can an agent never adopt? (immunity, made precise)

In the linear variant, immunity is $\tau_i^* > 1$, i.e.

$$q_i > \Gamma + \lambda. \tag{15}$$

Since $q_i \leq 1$, this region is **non-empty only when** $\lambda < 1 - \Gamma$: with $\lambda \geq 1$ immunity is impossible (a fully-adopted similar neighborhood can compensate any deficit). So the intuition "$q_i, \Gamma \in [0, 1]$ makes this very unlikely" is correct *at large* λ but not in general. Empirically, computed directly from $q_i =$ `mur_score` on the 96 GSS networks (the same 1000 agents in every network, so the shares do not vary across networks; GSS mean $q = 0.585$; isolates 0.11%), share of agents with $q_i > \Gamma + \lambda$:

λ	$\Gamma = 0$	0.1	0.2	0.3	0.4	average over the IUL grid
0.5	66.4%	46.5%	32.2%	16.1%	6.2%	14.0%
0.6	46.5%	32.2%	16.1%	6.2%	0	8.1%
0.7	32.2%	16.1%	6.2%	0	0	4.1%
0.8	16.1%	6.2%	0	0	0	1.7%
0.9	6.2%	0	0	0	0	0.4%
1.0	0	0	0	0	0	0

The share depends only on $\Gamma + \lambda$: 66.4, 46.5, 32.2, 16.1 and 6.2% for $\Gamma + \lambda = 0.5, \ldots, 0.9$, and zero from 1 on.

The immune share is only an **upper bound** on adoption, $\Phi \leq 1 - \Pr(q_i > \Gamma + \lambda)$ – isolates, and with the correct q the dynamics stay far below it at low IUL. At $\lambda = 0.5$ (random seeding) the highest adoption anywhere on the MSP axis is 3–4% for $\Gamma \leq 0.1$ (the seed cluster plus a few), 8% at $\Gamma = 0.2$ (bound 68%) and 27% at $\Gamma = 0.3$ (bound 84%), and jumps to 85–88% at $\Gamma = 0.4$ (bound 94%): most non-immune agents still need a large adopting share of their neighborhood ($\tau_i^* = (q_i - \Gamma)/\lambda$ close to 1), and the cascade stalls before the immune tail binds. The bound is a prediction, not a bug: *when social proof is weak, its reach is bounded.* The current engine cannot produce it because τ is truncated to $[0, 1]$ by construction. (Under the inverted q the immune tail was 10.9% at $\lambda = 0.5$, $\Gamma = 0.2$, and showed as a visible ceiling $\Phi \approx 0.75$.)

In the γ-form, immunity is instead $c_i^* > k_i$, i.e.

$$\left(\frac{q_i}{\Gamma}\right)^{1/\gamma} > k_i \quad \Longleftrightarrow \quad q_i \; > \; \Gamma\, k_i^{\gamma}, \tag{16}$$

which depends on the agent's **degree**: the out-of-reach agents are those both highly averse *and* poorly connected. Arguably more realistic — and a distinctive, testable difference between the two variants.

5.2 The log-space bridge: $\ln a$ vs. $\tilde{E}$

In log space, Tur's reinforcement variable is $\ln a$ — the log of the gated *count*. Is that "just" our normalized exposure? Almost: with $\tilde{E}_i = \tilde{A}_i / k_i$,

$$\ln \tilde{A}_i \; = \; \ln \tilde{E}_i \; + \; \ln k_i. \tag{17}$$

On **degree-regular substrates** — and Tur's Watts–Strogatz networks are degree-homogeneous with $k = 4$ — the term $\ln k_i$ is one global constant, absorbable into Γ (or u_0): identifying the reinforcement with $\ln \tilde{E}$ is then exact up to recalibration. On NDFU's **heterogeneous** networks (degrees roughly 1–80), the two differ by the agent-specific offset $\gamma \ln k_i$:

- using $\tilde{E}$ says "reinforcement is a *share of your attention*";
- using $\tilde{A}$ says "every endorsement adds the same, whether you have 4 neighbors or 40" — at the same adopted *fraction*, a hub receives more reinforcement.

This is the count-vs-fraction fork of §4.6, seen from the logarithm: the analogy $\ln a \leftrightarrow \tilde{E}$ is complete *modulo degree*, and the choice between the two variants is precisely the choice of what to do with $\ln k_i$.

5.3 What γ is — and is not — in Tur 2018 and Tur 2024

Read against both PDFs, the epistemic status of γ is the following.

What their justification does have.

- An **axiomatic characterization** (2018, p. 3; restated in 2024): the updated requirement $f(q; a, \gamma) = q \cdot (1/a)^\gamma$ must satisfy (a) decreasing in a for $\gamma > 0$; (b) decreasing in γ for $a > 1$; (c) $\gamma = 0$ replicates the percolation benchmark; (d) $a = 1$ gives $f = q$ (the *first* adopting neighbor only informs; reinforcement starts with the second — a clean echo of complex contagion's "more than one source").
- A **concavity argument** for the restriction $\gamma \leq 1$: "assuming that the marginal effect of each additional neighbor on the total social influence cannot be increasing" (2018), with the reductio that a very large γ trivializes diffusion *on their degree-4 lattices*.
- An interpretable **marginal effect** (2018, Eq. 2): $\Delta q_t^i = -\,q_0^i \gamma \,/\, (a_t^i)^{\gamma+1} \cdot \Delta a$ — the first adopting friends produce the large drops, and the drop is *proportional to the agent's own requirement* (more averse agents receive larger absolute reductions). This proportionality is the seed of the complementarity discussed in §5.5.

What it does not have.

- **No empirical calibration.** In both papers γ is a free "intensity" parameter, introduced with one sentence (2024: "for some innovations, social reinforcement may be more intense than for others … so we introduce social reinforcement as a parameter") and *swept*: $\gamma \in [0, 1]$ continuously in the 2018 phase maps and the 2024 heat maps, with focal values $\{0, 0.5, 1\}$ in 2024's Fig. 5 ("$\gamma = 1$ is maximum social reinforcement"). There are no units, no measurement strategy, no experimental anchor — not even a link to Centola's dose–response evidence, which would have been the natural one.
- **No transportability across degree scales.** All Tur simulations live on $k = 4$ neighborhoods, so $a \in \{1, \ldots, 4\}$ in practice and the reinforcement spans at most a factor $4^\gamma \leq 4$. On NDFU's imputed networks (mean degree ≈ 29, hubs beyond 60), the same γ produces reductions up to 29^γ: at $\gamma = 1$ a saturated average neighborhood divides the requirement by ≈ 29, a force of a different order of magnitude. Even their $\gamma \leq 1$ bound is reasoned *for* $k = 4$. **Borrowing Tur's γ values would inherit the symbol, not the calibration.**

Verdict. Adopting the γ-form buys (i) exact formal nesting, (ii) the published axioms and the concavity argument for free, and (iii) a citation in the target journal. It does *not* buy empirical values for γ, nor freedom from arguing the coupling's magnitude — if the γ-form is used, γ must be calibrated for *our* degree distribution (e.g. with the same fit-to-current-model procedure used for λ). Epistemically, γ and λ are in the same place: free couplings — with the difference that λ has the stronger microeconomic backing (§5.6).

5.4 Exposure traditions: the power form is wedded to counts

The γ-form cannot be written with normalized exposure, and the reason is structural, not stylistic: the multiplier $\tilde{A}^\gamma$ amplifies only because $\tilde{A} \geq 1$. Substituting the fraction $\tilde{E} \in [0, 1]$ gives $\tilde{E}^\gamma \leq 1$ — social proof would *reduce* situational utility. There is no natural "fractional γ-form".

Consequences:

- Choosing the γ-form moves the model from the fractional-threshold tradition (Granovetter 1978; Valente 1996; Watts 2002) — the tradition NDFU and `netdiffuseR` live in — to the absolute-count tradition (Centola & Macy 2007). For a *Social Networks* audience this is not neutral.
- **Mitigation 1 (mechanism ≠ measurement):** even if the decision rule runs on counts, Valente-style exposure-at-adoption can still be computed and reported *ex post* as a descriptive statistic (measured thresholds). Nothing forbids reporting $\tilde{E}_i$ at the moment of adoption.

- **Mitigation 2 (tooling):** `netdiffuseR` supports non-normalized exposure (`exposure()` with `normalized = FALSE`), so the count threshold $\lceil (q_i/\Gamma)^{1/\gamma} \rceil$ runs on the same machinery. The loss is conceptual continuity, not implementation.

5.5 Are preferences modified? Composition, not units

Formally, the γ-form does *not* force the "preferences change" reading: $\Gamma \cdot \tilde{A}^{\gamma} \geq q_i$ is symmetric, and the factor $\tilde{A}^{\gamma}$ can be booked on either side —

$$\underbrace{U_i(t) = \Gamma \cdot \tilde{A}_i^{\gamma} \geq q_i}_{\text{situational utility multiplied; preference fixed}} \qquad \Longleftrightarrow \qquad \underbrace{\Gamma \geq q_i \cdot \tilde{A}_i^{-\gamma} = q_{i,t}}_{\text{requirement lowered (Tur's framing)}} \tag{18}$$

with identical behavior. In logs the first reading is a Cobb–Douglas-style utility in two "goods" (product quality, social proof): $\ln U_i = \ln \Gamma + \gamma \ln \tilde{A}_i$. Both sides live on the same $[0, 1]$ utility scale as q_i, so the real distinction is not one of *units* but of **composition**: additive vs. multiplicative.

And the composition carries substance. Compare the marginal value of social proof:

$$\text{linear:} \qquad U_i = \Gamma + \lambda \tilde{E}_i, \quad \frac{\partial U_i}{\partial \tilde{E}_i} = \lambda, \qquad \frac{\partial^2 U_i}{\partial \Gamma \, \partial \tilde{E}_i} = 0 \qquad (\textbf{substitutes}), \tag{19}$$

$$\gamma\text{-form:} \qquad U_i = \Gamma \cdot \tilde{A}_i^{\gamma}, \quad \frac{\partial U_i}{\partial \tilde{A}_i} = \gamma \, \Gamma \, \tilde{A}_i^{\gamma - 1}, \quad \frac{\partial^2 U_i}{\partial \Gamma \, \partial \tilde{A}_i} = \gamma \tilde{A}_i^{\gamma - 1} > 0 \quad (\textbf{complements}). \tag{20}$$

In the linear form, a given (similar) adopting neighborhood contributes the same $\lambda \tilde{E}$ regardless of how good the product is: quality and social proof are *substitutes* (additively separable), and the social contribution needed to convert agent i compensates the utility deficit one-for-one, $\lambda \tilde{E}_i \geq q_i - \Gamma$, independently of Γ's level. In the γ-form the cross-partial is strictly positive (complementarity in the Edgeworth–Pareto sense): *the same adopting neighborhood is worth more for better products.* The mirror statement on the requirement side is Tur 2018's own Eq. 2, where the drop is proportional to q_0^i. The required social proof falls steeply with quality,

$$\ln c_i^* \;=\; \frac{\ln q_i - \ln \Gamma}{\gamma}, \qquad \frac{d \ln c_i^*}{d \ln \Gamma} \;=\; -\frac{1}{\gamma}, \tag{21}$$

an elasticity of $-1/\gamma$ — e.g. at $\gamma = 0.5$, doubling the product's quality divides the required count of similar adopters by four.

So the honest summary of the user's point: it *can* be framed as preferences-fixed either way, but the multiplicative interaction is a substantive behavioral commitment (quality–social-proof complementarity), not just bookkeeping — and since Tur themselves present the form as requirement-lowering, adopting their exact form and citation invites the "preferences change" reading unless a paragraph defends the other bookkeeping.

5.6 The Brock–Durlauf microfoundation of the linear form

The linear variant is not ad hoc: it is the deterministic limit of the canonical discrete-choice-with-social-interactions model of Brock & Durlauf (2001, *Review of Economic Studies* 68(2):235–260). In their setup, agent i chooses a binary action and derives utility

$$V_i(\omega_i) \;=\; \underbrace{h_i(\omega_i)}_{\text{private utility}} \;+\; \underbrace{J \, \omega_i \, \bar{m}_i^e}_{\text{social utility (conformity)}} \;+\; \underbrace{\varepsilon_i(\omega_i)}_{\text{random shock}}, \tag{22}$$

where $\bar{m}_i^e$ is the expected mean choice of i's *reference group* and $J \geq 0$ is the social-interaction (conformity) coefficient. Three points of contact:

1. **Additive separability of private and social utility is the standard assumption** of that literature — exactly the structure of the linear rule. Mapping: the private net utility of adopting is $\Gamma - q_i$; the social term is $\lambda \tilde{E}_i$, with $\tilde{E}_i$ playing $\bar{m}_i^e$ (the adoption rate of the reference group) and λ playing J. Adoption when total utility is non-negative: $\Gamma - q_i + \lambda \tilde{E}_i \geq 0$ is (3) rearranged.

2. **NDFU's additions are well-defined deviations, not violations:** the reference group is not exogenous but *similarity-gated* (the $\sigma(h - d_{ij})$ filter selects who belongs to i's effective reference group — our contribution), and the shock is degenerate ($\varepsilon \to 0$), turning Brock–Durlauf's logit choice probabilities into a deterministic threshold rule. Reintroducing a small ε would give a smooth (logit) adoption rule — a natural robustness variant.

3. **J has an econometric life that γ lacks.** The Brock–Durlauf coefficient underpins the social-multiplier / social-interactions estimation literature (see Blume, Brock, Durlauf & Ioannides 2011, *Handbook of Social Economics*, for identification), so λ inherits a path toward empirical anchoring. As a bonus, Brock & Durlauf themselves develop the mean-field correspondence of (22) with the Ising model — which connects this microfoundation directly to NDFU's criticality analysis.

5.7 Verdict table and recommendation

	Linear-fractional — $\Gamma + \lambda \tilde{E}_i \geq q_i$	**γ-form (count)** — $\Gamma \cdot \tilde{A}_i^{\gamma} \geq q_i$
Answer to Bianchi	✓ equally bottom-up (threshold derived from measured q)	✓ equally bottom-up
Backing of the coupling	λ with Brock–Durlauf microfoundation (additive private + social utility; §5.6); its magnitude still needs arguing	γ with Tur precedent (axioms + concavity), but **no empirical calibration and no transportability** across degree scales (§5.3)
Nesting in Tur 2024	only qualitative	**exact, on any graph** (gate off; ξ=Γ, u_0=q, same γ)
Exposure tradition	✓ Valente/Granovetter fractional; `netdiffuseR` native	counts (Centola–Macy/Watts); Valente exposure only as ex-post reporting (§5.4)
Preferences-fixed framing	✓ clean ("situational utility rises"; Brock–Durlauf)	defensible, but against the grain of the very citation it borrows (§5.5)
Quality × social proof	substitutes (social contribution independent of Γ)	**complements** ($\partial^2 U / \partial \Gamma \, \partial \tilde{A} > 0$) — strong assumption, but interesting and testable
Immunity	$q_i > \Gamma + \lambda$, degree-independent (§5.1)	$q_i > \Gamma k_i^{\gamma}$, **degree-dependent** (averse + poorly connected) — arguably more realistic
Empirical status	✓ **already validated**: replicates the current model's heatmaps; $\lambda \approx 0.7$ identified unimodally	not yet run; γ to be calibrated with the same procedure
Hub behavior	bounded ($\tilde{E} \leq 1$)	at high γ, a saturated hub receives reinforcement $\sim k^{\gamma}$ — may over-fire on dense networks

Recommendation. Use the **linear-fractional variant as the primary operationalization** (validated, Valente-compatible, microeconomically clean, λ identified from the legacy data), and the **γ-form as the formal bridge to Tur and as a robustness check**: present the exact nesting (§4.6) in the text, and run a short γ sweep — calibrated with our own fit procedure, never by borrowing Tur's numbers — to show that the conclusions do not hinge on the additive-vs-multiplicative composition. This keeps the best of both: the tradition and the validation of the linear form, and the exact nesting as an argument, without inheriting γ's transportability trap.

6 Side-by-side with the current model and Tur

	NDFU current	Tur 2024	**Unified rule (both variants)**
Decision rule	two channels (OR)	single utility rule	**single utility rule**
Threshold source	exogenous $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$	emerges from reinforcement	**derived from measured** q_i: $(q_i - \Gamma)/\lambda$ or $(q_i/\Gamma)^{1/\gamma}$
Preferences modified during diffusion?	no	yes (u lowered)	**no** under the situational-utility framing (§4.2, §5.5)
Distance enters via	stochastic gate in $\tilde{E}$	— (absent)	**same stochastic gate**, in $\tilde{E}$ or in $\tilde{A}$
Heterogeneity source	q empirical + τ artificial	u artificial	q **empirical only**
Free params (beyond Γ, h)	τ_{mean}, τ_{sd}	γ	λ (linear) or γ (γ-form)
Nests Tur 2024?	no	—	**exactly via the γ-form** (§4.6); linear form only qualitatively
`netdiffuseR` as-is?	yes	no	**yes** (fractional or count `threshold.dist`)
Main risk	Bianchi critique	distance-blind	heatmaps shift with the τ–q coupling (checked: they don't, §7)

7 Calibration notes and empirical status

- Comparability anchor: the main-text configuration of the current engine is $\tau \sim \mathcal{N}(0.40, 0.12)$. Under the linear-fractional variant, $\tau_i^* = \max\{0, (q_i - \Gamma)/\lambda\}$, so the implied threshold distribution *depends on* Γ *and the empirical* q *distribution.* The sweep script reports, per (λ, Γ): mean and SD of τ^*, share of instigators ($\tau^* = 0$), and share of immunes ($\tau^* > 1$).
- **Empirical status (linear variant).** An initial minimal test (12 networks, coarse grid; script since removed) showed the qualitative phase structure survives. The extended sweep (`10_unified_model_sweep.R`: 96 GSS networks, the engine's full 41×13 grid, $\lambda \in \{0.3, 0.4, \ldots, 1.5\}$, random seeding, 665,184 simulations) found a clean **unimodal best fit to the current model at** $\lambda \approx 0.7$ ($r = 0.94$ total, 0.98 rational, 0.93 social against the $\tau \sim \mathcal{N}(0.40, 0.12)$ surface; $\lambda \in [0.6, 0.9]$ all $r > 0.91$). Known differences: tipping one MSP-grid step earlier, and the immunity ceiling at low λ (§5.1).
- On the λ range: $\{0.3, \ldots, 1.5\}$ was a pragmatic bracketing (the minimal test showed 0.5 and 1.0 straddling the current model and 2.0 clearly too strong); the sweep's unimodal peak with degradation at *both* flanks justifies the endpoints ex post. Theoretical landmark: $\lambda = 1$ is the **immunity boundary** — since $q \leq 1$, $\lambda \geq 1$ implies zero structurally-immune agents at every $\Gamma > 0$, while $\lambda < 1$ leaves an immune share $\Pr(q > \Gamma + \lambda)$. There is no theoretical bound at $\lambda = 1$ (λ is a utility contribution, like Brock–Durlauf's unbounded J); production runs use the calibrated band $[0.5, 0.9]$.
- Still pending: the mirror γ-form sweep (count threshold $\lceil (q_i/\Gamma)^{1/\gamma} \rceil$ on the same machinery, γ calibrated by the same fit procedure), the remaining seeding strategies, and the ATP/innovation system.

7.1 The structural premium survives the unified rule

Run by `scripts/05_unified_diffusion_sweep.R`: GSS vs. genuine degree-preserving twins (`igraph::rewire(keeping_degseq)`, built in memory with the recipe of `scripts/04_make_DP_networks.R`), $\lambda \in \{0.5, \ldots, 1.0\}$, 41 IUL levels × a 25-level MSP

grid (step 1/24, nesting the legacy 13-level grid), 96 networks, five seeding strategies (1,180,800 simulations each), with q_i = `mur_score` = 1 − propensity$_i$ (GSS mean 0.585). Analysis in `scripts/06_premium_sensitivity.R` (λ profile, link-free diagnostics) and `scripts/06b_premium_bootstrap.R` (joint BAM of the strategies and run-level bootstrap; all intervals below).

Seed matching. The primary seed is drawn once per run, independently of λ. For *random*, *marginal* and *central* it is the *same node* in a GSS network and its DP twin; *closeness* and *eigen* use each topology's own most central node. The seed cluster is the primary plus its nearest nodes, 0.40× the network's maximum degree in total ($\approx$ 28 seeds). A network and its twin share the degree sequence, so the cluster has the same size in both (checked in all 96 runs of every strategy), and the strategies differ only in where they seed. With a low-degree primary the cluster reaches into the second neighborhood, so its shape differs too, not only its center. The *random* strategy is the reference.

Uncertainty: the run is the unit. The 1025 cells of a run share the GSS network, its DP twin and the primary draw, so they are not independent observations: the unit of replication is the **run** (96 per topology). Run k also uses the same network, twin and primary draw in every strategy, so contrasts between strategies are **paired**. All intervals come from a run-level bootstrap: resample `run_id` (1..96) with replacement, with one set of weights shared by GSS, DP and every strategy, and refit ($B = 1000$; 95% percentile intervals). The model, per λ, is a joint BAM

$$\operatorname{logit}\Phi \;=\; \alpha_s + \beta_{\mathrm{DP},s}\,\mathbf{1}_{\mathrm{DP}} + f_s(\Gamma, h), \qquad f_s = \texttt{te}(\Gamma, h, k = 12 \times 12) \text{ per strategy } s,$$

fitted on run-aggregated counts (exact: the model has no run term). Three choices, decided by checking them against the data:

- **The surface varies by strategy.** With one shared surface, each strategy's misfit leaks into the strategy × topology terms and flattens the contrasts (at $\lambda = 1$, central − random shrinks from −0.21 to −0.08). With strategy-specific surfaces the joint model factorizes and reproduces the separate per-strategy fits exactly; what it adds is the paired bootstrap of the contrasts.
- **It does not vary by topology.** β_{DP} is *defined* as the constant logit shift between GSS and DP. A topology-specific surface would absorb the premium itself. The premium is really a shift of the tipping frontier, and its shape is described by the link-free gap and the MSP_c shift, not by the BAM.
- **Basis 12×12, not $k = 10$.** The $s(\Gamma, h, k = 10)$ surface of 05/06 is too stiff for the sharp frontier (deviance explained 0.94–0.99). β_{DP} converges at a 12×12 tensor basis; $k = 10$ understates $|\beta|$ by 3–13%, more at high λ. Contrasts between strategies barely move. $k = 10$ values are kept for continuity.

The pooled premium is the unweighted mean of the six per-λ estimates ("λ-mean"). The pooled fit of 05 has one intercept per λ but puts one surface under all of them, and is attenuated (random: −0.718 vs. −0.763 with the same $k = 10$ basis; an earlier version also left out the λ intercepts and gave −0.684).

λ	β_{DP} (odds) te 12×12	odds drop	β_{DP}, $k = 10$ (as in 05)	$\bar{\Phi}_{\mathrm{GSS}} - \bar{\Phi}_{\mathrm{DP}}$ (pp)	% of plane with gap > 0.10	mean gap in that band (pp)
0.5	-0.499 $[-0.540, -0.434]$	-39%	-0.482	3.98 $[3.86, 4.05]$	18.1	16.5
0.6	-0.630 $[-0.692, -0.536]$	-47%	-0.589	4.52 $[4.39, 4.60]$	20.3	17.6
0.7	-0.834 $[-0.934, -0.689]$	-57%	-0.753	5.46 $[5.29, 5.58]$	20.8	21.3
0.8	-0.971 $[-1.101, -0.787]$	-62%	-0.851	5.95 $[5.78, 6.08]$	20.8	23.4
0.9	-1.054 $[-1.216, -0.837]$	-65%	-0.922	6.30 $[6.11, 6.44]$	21.2	24.6
1.0	-1.118 $[-1.306, -0.881]$	-67%	-0.980	6.57 $[6.37, 6.72]$	21.5	25.8
mean	-0.851 $[-0.963, -0.697]$	-57%	-0.763	5.46 $[5.30, 5.57]$		

Random seeding; 95% run-bootstrap intervals below each estimate. The last two columns come from 06 and have no intervals yet. MSP_c, GSS $\to$ DP: .085$\to$.156, .088$\to$.157, .082$\to$.153, .079$\to$.151, .078$\to$.147 and .077$\to$.148 for $\lambda = 0.5, \ldots, 1.0$.

Reading. The premium survives the bottom-up engine across the whole λ band and for all five seeding strategies, and no interval comes near zero: rewiring the ties lowers the odds of adoption by 39–67% under random seeding (λ-mean $\beta_{\mathrm{DP}} = -0.851$ $[-0.963, -0.697]$, -57%; probability scale 5.46 pp $[5.30, 5.57]$). It grows smoothly and monotonically with λ (steps per decile -0.131, -0.203, -0.137, -0.083, -0.063): the stronger social proof is, the more the real fabric matters. The honest intervals are 7–18 times the naive BAM standard errors for the hub strategies and 64–220 times for random and marginal, whose primary seed is a lottery.

λ is a free coupling, not a calibrated constant, so the paper should report results as functions of λ. Two scales must be read together:

- on the **odds** scale the premium more than doubles over the band ($-0.50 \to -1.12$; $-0.48 \to -0.98$ with the $k = 10$ surface);
- on the **probability** scale it grows by $+65\%$ (4.0 $\to$ 6.6 pp), with much tighter intervals and no dependence on the surface basis.

Part of the difference is a link effect (cells with $\Phi > 0.95$ go from 40% to 61% of the plane for GSS, from 32% to 48% for DP, and the logit inflates a given probability gap there), but most of the growth is genuine structural separation.

The λ-robust statements (recommended for the paper):

1. **Where it lives.** The premium is concentrated in an L-shaped band along the tipping frontier: 18–22% of the (IUL, MSP) plane, where GSS gains 16–26 percentage points. Outside that band the two topologies agree (both dead, or both saturated). Structure decides where the system is undecided.
2. **Tipping shift.** DP needs 1.8–1.9 times the social openness to tip ($MSP_c \approx 0.077$–0.088 for GSS vs. 0.147–0.157 for DP). The absolute shift is the most stable way to state the premium: 0.069–0.071 across the band under random seeding ($\pm 2\%$), and 0.060–0.076 across all five strategies.
3. **Channel split** (06, $k = 10$ surface). The rational channel is flat in λ ($\beta \approx -0.146$) while the social channel carries the growth ($-0.21 \to -0.37$). The rational channel is *not* topology-blind: its adopters still need one gate-passing contact, and DP ties are socially more distant (mean Gower $0.229 \to 0.325$), so the gate rejects more often.

Seeding strategies. λ-mean and endpoint β_{DP} (te 12×12), the λ-mean probability-scale gap and the tipping shift, by strategy (mean seed-cluster size in parentheses; 95% run-bootstrap intervals):

strategy	β_{DP} λ-mean	β_{DP} $\lambda = 0.5$	β_{DP} $\lambda = 1.0$	$k = 10$ λ-mean	gap (pp) λ-mean	MSP_c shift
central (28.7)	-0.965 $[-0.976, -0.954]$	-0.533 $[-0.540, -0.526]$	-1.325 $[-1.344, -1.304]$	-0.877	5.61 [5.55, 5.68]	.060–.068
closeness (28.7)	-0.961 $[-0.973, -0.951]$	-0.535 $[-0.542, -0.529]$	-1.320 $[-1.340, -1.299]$	-0.874	5.60 [5.54, 5.66]	.062–.074
eigen (28.7)	-0.949 $[-0.961, -0.936]$	-0.523 $[-0.529, -0.516]$	-1.319 $[-1.343, -1.295]$	-0.864	5.55 [5.47, 5.62]	.065–.076
random (28.4)	-0.851 $[-0.963, -0.697]$	-0.499 $[-0.540, -0.434]$	-1.118 $[-1.306, -0.881]$	-0.763	5.46 [5.30, 5.57]	.069–.071
marginal (28.4)	-0.852 $[-0.965, -0.700]$	-0.508 $[-0.549, -0.445]$	-1.091 $[-1.282, -0.865]$	-0.762	5.44 [5.29, 5.54]	.068–.074

Paired contrasts (same run in both strategies). On β, a negative value means the first strategy has the *larger* premium; on the gap, a positive one does:

contrast	$\Delta\beta_{\mathrm{DP}}$, λ-mean	$\lambda = 0.5$	$\lambda = 1.0$	Δ gap (pp), λ-mean
central − random	-0.114 $[-0.271, -0.002]$	-0.034 $[-0.098, 0.008]$	-0.207 $[-0.448, -0.014]$	0.15 [0.05, 0.31]
closeness − random	-0.110 $[-0.267, 0.003]$	-0.036 $[-0.101, 0.005]$	-0.202 $[-0.440, -0.009]$	0.14 [0.03, 0.30]
eigen − random	-0.098 $[-0.252, 0.017]$	-0.024 $[-0.089, 0.018]$	-0.201 $[-0.441, -0.006]$	0.08 [−0.04, 0.24]
random − marginal	0.001 [−0.246, 0.249]	0.009 [−0.086, 0.106]	-0.026 [−0.404, 0.362]	0.02 [−0.15, 0.20]
central − closeness	-0.004 $[-0.013, 0.006]$	0.002 [−0.003, 0.008]	-0.005 [−0.022, 0.011]	0.01 [−0.04, 0.07]
central − eigen	-0.016 $[-0.026, -0.005]$	-0.010 $[-0.017, -0.004]$	-0.006 [−0.026, 0.012]	0.07 [0.01, 0.13]
closeness − eigen	-0.012 $[-0.023, -0.002]$	-0.012 $[-0.019, -0.006]$	-0.001 [−0.022, 0.021]	0.06 [−0.01, 0.12]

- **The three hub strategies are interchangeable.** Central and closeness do not differ ($\Delta\beta = -0.004$ $[-0.013, 0.006]$); eigen's premium is distinguishably smaller, but by 1.6% of β (0.07 pp). The hub premium is also the most precisely estimated (interval width ≈ 0.02 in β): the primary is fixed by the network, so runs differ only in network and dynamics.
- **Random and marginal coincide** ($\Delta\beta = 0.001$ $[-0.246, 0.249]$; gap 0.02 pp $[-0.15, 0.20]$): a typical and a peripheral primary seed equally well. Their premium is ten times less precise than the hubs' (width ≈ 0.27): where the primary lands is a lottery.
- **The hubs add a small premium, borderline on the odds scale.** Central − random is -0.114 $[-0.271, -0.002]$ on the λ-mean and 0.15 pp $[0.05, 0.31]$ on the gap (3% of the 5.5-pp premium; t interval $[0.02, 0.28]$). It grows with λ: indistinguishable from zero at $\lambda = 0.5$ and 0.7–0.8, distinguishable at $\lambda \geq 0.9$ (-0.207 $[-0.448, -0.014]$ at $\lambda = 1$; 0.31 pp $[0.12, 0.52]$) and, barely, at $\lambda = 0.6$. Closeness behaves like central (-0.110 $[-0.267, 0.003]$; 0.14 pp $[0.03, 0.30]$). Eigen's margin is the smallest and not distinguishable on the λ-mean (-0.098 $[-0.252, 0.017]$; 0.08 pp $[-0.04, 0.24]$), but at $\lambda = 1$ all three hubs are ($\Delta\beta \approx -0.20$; 0.29–0.33 pp).

A reverse-premium corner (conditional, not universal). In a small region — high openness ($h \gtrsim 0.6$, median 0.83) and moderate attractiveness ($\Gamma \approx 0.275$–0.45) — **DP beats GSS**. To

tell it from noise, each cell gets the paired z of its gap over the 96 runs, and counts as a **robust DP win** only if it clears a one-sided *simultaneous* threshold: max-T over the 1025 cells of its strategy $\times$ λ panel, familywise 5%, from the same run bootstrap ($z < -3.6$ to -3.8, against -2 pointwise; `premium_reverse_corner.csv`):

- **Robust at $\lambda = 0.5$–0.7, identically in all five strategies.** 47–48, 59–61 and 38 corner cells clear the threshold, including *every* cell where DP wins by more than 2 pp (2.6–3.4%, 2.4–2.6% and 0.8% of the plane, up to 7.0, 5.8 and 3.0 pp). At $\lambda = 0.5$ it spills just past the box ($\Gamma = 0.475$–0.525, ≤ 1 pp).
- **Detectable but negligible at $\lambda = 0.8$** (12–13 corner cells, ≤ 0.14 pp, plus 8 cells at $\Gamma = 0.25$ with ≤ 0.35 pp); at $\lambda \geq 0.9$ at most 0.04 pp remains.
- **The low-IUL strip** ($\Gamma \leq 0.1$, DP ahead by ≤ 0.05 pp) is noise for the hub strategies and random: no cell clears the threshold (up to 19 pass pointwise). For marginal it is systematic (30–48 cells per λ, $h \gtrsim 0.5$) but negligible (≤ 0.14 pp): an effect of the seed cluster's geometry, not a reversal of the premium.

The corner tracks the immunity boundary. Its strong part (> 2 pp) lies at $\Gamma + \lambda \approx 0.85$–$0.975$, where a small immune tail remains (13 to 1% of agents), and the robust cells drift to lower Γ as λ grows ($\Gamma = 0.35$–0.45 at $\lambda = 0.5$, 0.275–0.375 at 0.7, 0.275–0.3 at 0.8). There the most averse non-immune agents need nearly all of their neighbors to adopt (τ_i^* close to 1). The mechanism is Tur (2024)'s own: when the gate is wide open, homophily *isolates the resistant among the resistant*, whereas random mixing seats adopters next to reluctant agents and converts them. Once influence stops being similarity-gated, the sign of the structural effect can flip (the similarity-vs-diversity question). This is a lead, not an established mechanism (it did not hold under the inverted q). The corner keeps the premium claim properly conditional, but it is a weak-coupling feature and should not be presented as present at every λ.

Immunity under the correct q. Agents with $q_i > \Gamma + \lambda$ can never adopt. At $\Gamma = 0$ they are 66, 47, 32, 16, 6 and 0% of agents for $\lambda = 0.5, \ldots, 1.0$; averaged over the IUL grid, 14.0, 8.1, 4.1, 1.7, 0.4 and 0%. Mean adoption over the plane is correspondingly lower (random seeding: $\bar{\Phi}_{\mathrm{GSS}} = 0.61$–$0.74$, $\bar{\Phi}_{\mathrm{DP}} = 0.57$–$0.68$).

What changed with respect to the inverted q. Until 2026-10-05 the engine read the propensity score itself as q_i (high = *more* willing), the opposite of its role as a minimum utility requirement (GSS mean 0.415 instead of 0.585); the same inversion affected the legacy engine. With the correct q: (i) the premium survives in every strategy and keeps its shape (L-shaped band, flat rational channel, stable tipping shift of ≈ 0.065–0.07); (ii) it is smaller on the odds scale (λ-mean with $k = 10$: central $-1.07 \to -0.88$, random $-0.81 \to -0.76$) and, relative to its level, steeper in λ (β at $\lambda = 1$ over $\lambda = 0.5$: central $2.0 \to 2.3$, random $1.6 \to 2.0$); (iii) the reverse corner moves to higher Γ (from 0.10–0.275, by about the 0.17 shift in mean q) and is no longer present at every λ; (iv) the immune share rises 2.7-fold or more (IUL-grid average $5.2 \to 14.0$% at $\lambda = 0.5$). The inverted-q outputs are archived in `output/**/_qinverted/` (closeness and eigen references: the pre-seed-fix runs in `_prefix/`, which match the seed-matched code for the deterministic strategies: central -0.926 vs. -0.928). Those runs predate the common seed budget: except for central, whose cluster is the same under both rules, they seeded 0.40$\times$ the degree of the primary, so central is the like-for-like comparison.

Still open.

- **What the intervals cover.** The bootstrap treats the 96 networks as exchangeable draws from the fitted ERGM. They share the same 1000 imputed agents (the q composition is fixed), so the intervals cover tie formation, the seed draw and the dynamics, not survey sampling of the agents or uncertainty in the ERGM coefficients.

- **Specification.** The level of β_{DP} depends on the surface basis (3–13% between $k = 10$ and te 12×12). That is not in the bootstrap intervals; the contrasts are robust to it, and the link-free gap, which does not depend on it, should lead in the paper.
- **No intervals yet** for the band statistics, the channel split and the MSP_c shift of 06; the same bootstrap weights would give them.
- The legacy reference ($\beta = -0.850$) and the $\lambda \approx 0.7$ calibration (§7) were computed with the inverted q and are pending recomputation. (The immunity table of §5.1 is now recomputed.)

8 To verify before writing any of this into the paper

☒ Exact reinforcement functional form in Tur 2024: $u_t = u_0 \cdot (1/a_t)^{\gamma}$ (multiplicative, power-law). *Resolved against both PDFs (2026-08-31):* Tur 2018 (PRE), Eq. (1), uses the **same** form (there called MQR, $q_t^i = q_0^i(1/a_t^i)^{\gamma}$); $a_t^i \in \mathbb{N}$ is explicitly the number of **adopting neighbors** (a state count, not received recommendations), with the rule evaluated only once informed ($a \geq 1$). Both papers restrict $\gamma \in [0, 1]$ — justified in 2018 by non-increasing marginal social influence (concavity), with the reductio that large γ trivializes diffusion on their $k = 4$ lattices — and sweep γ as a free "intensity" parameter (2024, Fig. 5: $\gamma \in \{0, 0.5, 1\}$; heatmaps over $[0, 1]$). **No empirical calibration of γ exists in either paper**; see the transportability caveat in §5.3.

☐ Whether Tur 2024's homophily-rewiring ρ has a formal analogue we should cite when claiming nesting (their $\rho \approx$ our imputation; not identical constructs).

☐ The random-utility framing sentence for IUL (already drafted in `JOURNAL_FIT_ANALYSIS.md`, §4d) composes cleanly with the situational-utility framing: $U_i(t) = \Gamma + \lambda \tilde{E}_i$ is a private valuation plus a social term — this is Brock & Durlauf (2001)'s decomposition (§5.6), worth citing.